

Indian Institute of Information Technology, Guwahati

Road Network Segmentation from Satellite Images Using the SAM Model

B.Tech 6th Semester Project

Submitted by

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1 Introduction

Satellite imagery is widely used in urban planning, navigation, infrastructure monitoring, and disaster response. In many of these applications, the automatic extraction of road networks is an important prerequisite because roads provide the structural backbone for transportation analysis, accessibility estimation, and geographic information system updates. However, road extraction from satellite images remains challenging due to variations in road width, surface appearance, illumination, occlusion by trees or buildings, and the presence of visually similar background structures.

This project ultimately aims to generate accurate road masks from satellite imagery using the Segment Anything Model (SAM). The broader goal is to adapt SAM to the road-extraction setting so that it can predict road regions effectively and later support structured road-network analysis. To reach this objective in a systematic way, a U-Net-based segmentation model was first implemented as a baseline and learning stage. This intermediate stage helped build practical understanding of deep learning, image processing, segmentation training, and remote-sensing data before moving toward the final SAM-based objective. The project uses the DeepGlobe Road Extraction Dataset as the primary benchmark, with future extensions planned on CityScale/20Cities and SpaceNet RGB 1.0 meter imagery.

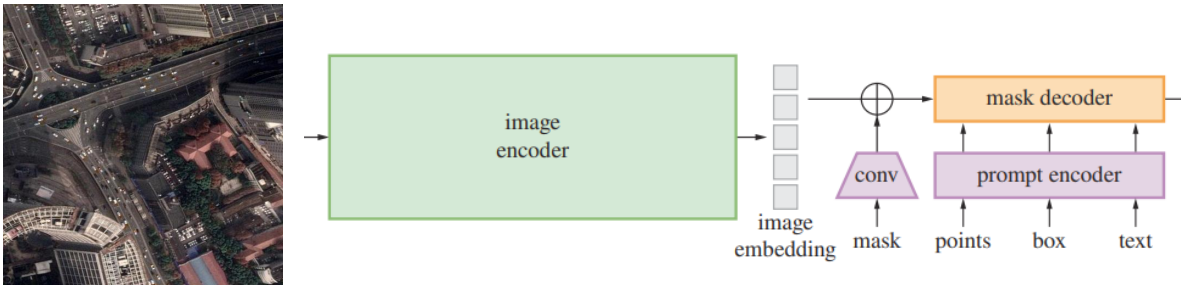


Figure 1: Road-segmentation workflow from an input satellite image to the Segment Anything Model pipeline, where the image is encoded and processed through the prompt encoder and mask decoder to generate segmentation masks.

2 Related Work

Road extraction has traditionally been studied as a semantic segmentation or boundary detection problem. Earlier approaches relied on classical image processing methods such as thresholding, edge detection, morphological operators, and hand-crafted linear feature extraction. Although these methods can work in constrained settings, they are often sensitive to shadows, noise, cluttered backgrounds, and changes in image quality.

With the growth of deep learning, convolutional encoder–decoder models such as U-Net and related segmentation architectures significantly improved performance on remote sensing imagery by learning road appearance directly from labeled data. More recent work has gone beyond pixel-wise masks and has aimed to recover full road graphs, where both geometry and topol-

ogy are explicitly modeled. This graph-based view is useful because it captures not only which pixels belong to roads, but also how intersections and road segments are connected.

The Segment Anything Model introduced a strong foundation-model approach for promptable image segmentation (Kirillov et al., 2023). Its large-scale pretraining makes it attractive for domains where labeled data are limited or where one wants to reduce training costs. In the road extraction context, SAM can be adapted to predict road and intersection masks, after which graph vertices can be extracted and linked. The proposed work is inspired by this line of thinking and also draws from recent road-network graph extraction ideas such as direct topology prediction for large-scale regions. The project is linked with the idea that visual embeddings from a foundation model can be combined with lightweight graph reasoning to produce efficient and scalable road network extraction.

3 Motivation

The motivation for this project comes from the growing importance of accurate road-network extraction from satellite imagery in both research and real-world applications. In the modern world, up-to-date road information is essential for navigation systems, smart-city planning, infrastructure monitoring, logistics, disaster-response coordination, rural accessibility analysis, and digital map updating. As satellite imagery becomes increasingly available, there is a strong need for intelligent systems that can automatically convert visual data into reliable geographic information. However, extracting roads automatically from satellite images remains difficult because roads are often thin, elongated, partially occluded, low-contrast, or visually similar to surrounding structures such as rooftops, riverbeds, and shadows.

This project is also highly relevant in the present technological landscape. It lies at the intersection of computer vision, remote sensing, and structured prediction, and it explores how a modern foundation model can be adapted to a domain-specific real-world problem. In an era where AI is increasingly expected to work across diverse applications with limited manual supervision, this project reflects an important research direction: using strong pretrained models to reduce annotation effort, improve generalization, and accelerate the development of practical intelligent systems.

A further motivation is the practical limitation of existing approaches. Traditional image-processing methods are often too fragile for complex real scenes, while many deep-learning methods require extensive task-specific design and large amounts of labeled data. This project is therefore motivated by the possibility of using SAM as a strong pretrained foundation model that may reduce the burden of manual annotation and domain-specific engineering while still producing useful road masks.

At the same time, the project has also been a learning process. Before directly attempting the final SAM-based objective, a U-Net-based segmentation pipeline was implemented to build a clear understanding of model training, image preprocessing, loss behaviour, mask prediction, and evaluation on satellite imagery. This baseline stage has been important in developing practical skills in deep learning, image processing, and the broader remote-sensing domain.

The project is also motivated by the need to move beyond pixel-level segmentation toward structured road-network understanding. In many practical settings, a binary road mask alone is not sufficient; one ultimately needs a connected graph representation that can describe intersections, links, and overall road topology. For this reason, the project aims not only to segment roads but also to create the basis for graph construction from the predicted outputs.

4 Methodology and Execution

The project follows a pipeline-based methodology. First, satellite images are taken from the DeepGlobe Road Extraction Dataset, which provides benchmark imagery and corresponding road labels. Additional experiments are planned on CityScale/20Cities and SpaceNet RGB 1.0 meter data in order to test the generalizability of the method across datasets.

The final target model in the project is SAM. The intended system is a SAM-based road-extraction framework that adapts the segmentation capability of SAM to the prediction of road regions and, later, road-network structure. However, before reaching that final objective, a U-Net-based segmentation pipeline was implemented as an intermediate baseline and learning stage. This made it possible to understand image preprocessing, segmentation training, mask prediction, and evaluation on remote-sensing imagery in a more controlled way. Based on this progression, the current methodology uses the U-Net stage as a practical foundation while the next major step is to adapt and strengthen the SAM-based pipeline for the final objective.

The implementation environment is Google Colab because dedicated college GPU access is currently unavailable. This keeps the system lightweight and practical while still allowing experimentation with pretrained models. Evaluation has already started, the final evaluation will likely include segmentation-based measures such as Intersection over Union (IoU), Precision, Recall, and F1-score, together with graph-level or topology-oriented assessment and inference time. Comparison will be made against baseline segmentation methods and, where feasible, against recent graph-extraction approaches reported in other research paper.

4.1 SAM Architecture

The proposed architecture is based on the Segment Anything Model and focuses on how image embeddings, prompt tokens, and decoder components interact to produce road masks. First, the satellite image is encoded into dense image embeddings, while prompt-related tokens are combined with output tokens to guide the segmentation process toward road structures of interest.

Inside the mask decoder, self-attention, token-to-image attention, image-to-token attention, and multilayer perceptron blocks iteratively exchange information between the learned image representation and the output tokens. This design allows the model to refine the correspondence between visual road patterns and the predicted segmentation output.

The final decoder stage generates road masks together with IoU confidence scores, which provide an estimate of mask quality. For this project, the same architecture is adapted to emphasize thin and connected road regions, making it suitable for road-segmentation and later

graph-construction tasks from satellite imagery.

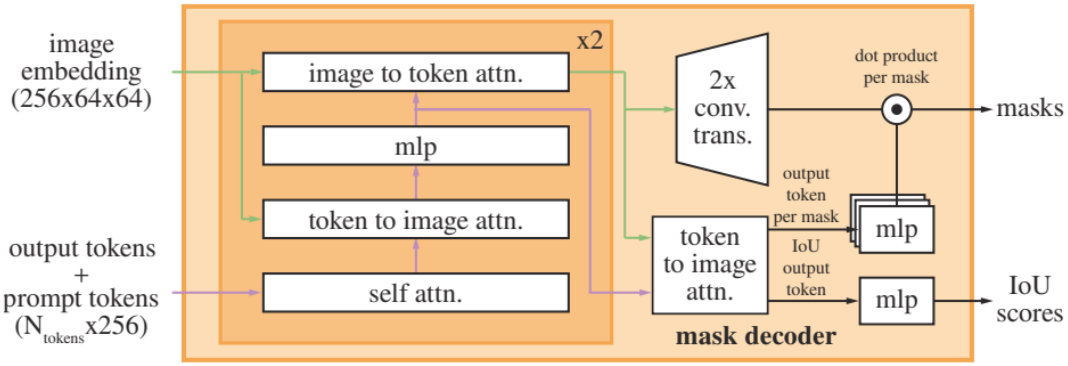


Figure 2: Architecture of the SAM mask decoder used as the basis for the proposed road-segmentation framework.

4.2 U-Net Architecture

As part of the baseline and learning stage of the project, a U-Net-based segmentation architecture was implemented for road-mask prediction. U-Net is well suited to this task because it combines precise localization with multiscale feature extraction, which is especially important for roads that appear as thin and elongated structures in satellite images. The architecture follows an encoder–decoder design, where the encoder progressively captures higher-level semantic information and the decoder reconstructs spatial detail to produce a dense segmentation map.

In the encoder path, the input image passes through repeated convolutional blocks followed by downsampling operations. These layers learn increasingly abstract visual features, beginning with low-level edge and texture information and gradually moving toward more semantic representations of road regions. As the feature depth increases, the spatial resolution is reduced, allowing the network to capture broader contextual information that helps distinguish roads from confusing background patterns.

At the bottleneck stage, the network contains the most compressed representation of the image, where high-level contextual understanding is strongest. In the decoder path, upsampling operations progressively restore spatial resolution. Skip connections transfer feature maps from the encoder to the decoder at corresponding scales, helping the model recover fine-grained boundary information that might otherwise be lost during downsampling. This is particularly valuable for preserving narrow road structures and improving the continuity of segmented road regions.

The proposed version shown below also includes attention-gate mechanisms. These gates help the network focus more strongly on relevant road features while suppressing less useful background responses. In practice, this can improve the quality of segmentation in visually complex scenes where shadows, buildings, vegetation, and other linear patterns may interfere with road extraction. For the present project, this architecture provides a strong and interpretable segmen-

tation backbone whose outputs can later support intersection detection and graph construction.

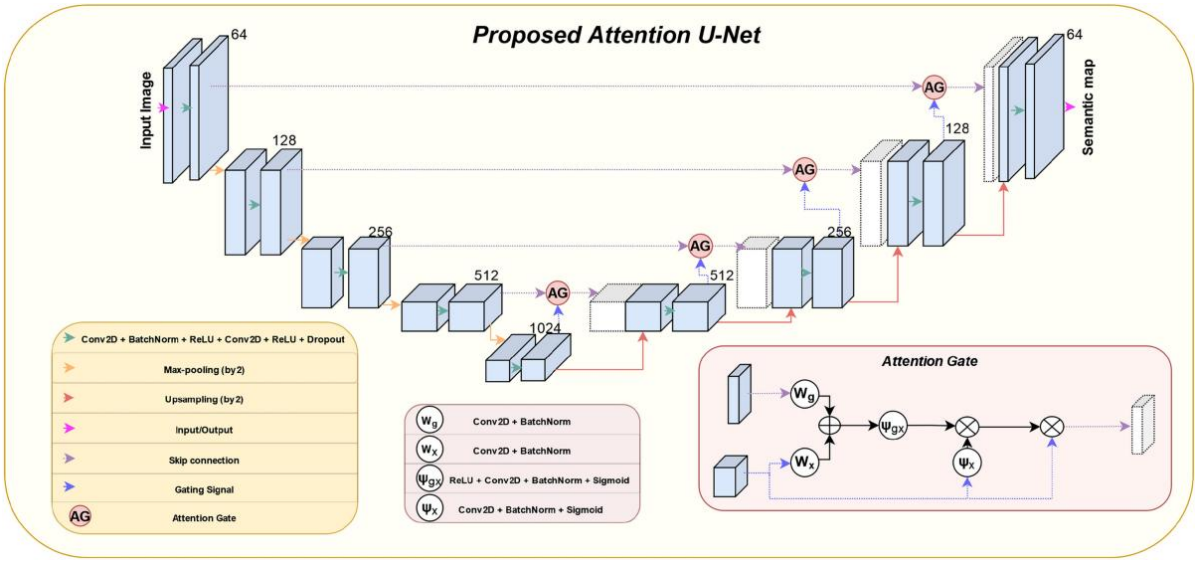


Figure 3: Attention U-Net architecture considered for the road-segmentation pipeline. The figure illustrates the encoder–decoder structure, skip connections, upsampling path, and attention gates used to refine relevant feature propagation.

5 Results

5.1 Datasets

The dataset used in this work is the DeepGlobe Road Extraction Dataset, which was released for the DeepGlobe road extraction challenge. This dataset is used as the main source for model development, validation, and result analysis in the present work. It can be accessed as the DeepGlobe Road Extraction Dataset on Kaggle.

A second dataset considered in this work is Satellite Images for Road Segmentation. This dataset is included as an additional resource for comparative road-segmentation experiments. It can be accessed as the Satellite Images for Road Segmentation dataset on Kaggle.

Dataset	Train	Validation	Test	Accuracy
DeepGlobe Road Extraction	6226	1243	1101	0.97
Satellite Images for Road Segmentation	5234	1103	509	0.94

Table 1: Dataset summary for the present work. It's train, test, validation and their corresponding Accuracy for 25 epochs.

5.2 Experimental Setup

The experiments in this project are carried out using the DeepGlobe Road Extraction Dataset as the primary benchmark for road-segmentation analysis. The implementation is developed in Python using deep-learning libraries within Google Colab, which serves as the main execution environment because dedicated institutional GPU resources are currently unavailable.

The current reported experiments correspond to the U-Net-based baseline stage of the project. In this supervised setup, satellite images are paired with corresponding road masks, and the dataset is divided into training, validation, and test splits so that learning progress and generalization can be monitored systematically. This stage is used to build practical experience with deep learning and image-processing workflows before moving to the final SAM-based objective. For the U-Net experiments, the current training schedule is limited to 25 epochs due to compute-time constraints associated with shared cloud GPU environments such as Google Colab and Kaggle.

The main evaluation focus is on segmentation quality and practical usability. Quantitative monitoring includes accuracy, Dice coefficient, and Intersection over Union (IoU), together with validation-side performance to understand how well the model generalizes beyond the training set. In addition to these numerical indicators, qualitative inspection of predicted road masks is used to examine road continuity, preservation of thin structures, and robustness in visually complex satellite scenes. This experimental setup provides a practical basis for comparing model behaviour, understanding the strengths and limitations of the baseline, and guiding the next stage of work toward a stronger SAM-based road-extraction pipeline.

5.3 Road Segmentation

The current results reported in this project correspond primarily to the U-Net-based baseline that was implemented before the final SAM-based objective. These results show that the baseline model is able to identify road regions from satellite images with reasonable visual consistency and that it provides a useful starting point for understanding segmentation behaviour in the road-extraction setting.

The comparison between the original image, the ground-truth mask, and the predicted mask indicates that the model is learning the primary road structure while suppressing much of the surrounding background.

To further support the present baseline results, the model summary of the U-Net architecture used in the road-segmentation pipeline is presented below. The summary shows the encoder and decoder stages, intermediate output dimensions, and trainable parameter counts across layers. This information helps document the internal structure of the segmentation network and clarifies the computational design currently used during this learning stage of road-mask prediction.

Model: "UNetModel"

Layer (type)	Output Shape	Param #	Connected to
InputLayer (InputLayer)	(None, 256, 256, 3)	0	-
EncoderLayer1 (EncoderBlock)	[(None, 128, 128, 32), (None, 256, 256, 32)]	10,156	InputLayer[0][0]
EncoderLayer2 (EncoderBlock)	[(None, 64, 64, 64), (None, 128, 128, 64)]	55,552	EncoderLayer1[0]…
EncoderLayer3 (EncoderBlock)	[(None, 32, 32, 128), (None, 64, 64, 128)]	221,696	EncoderLayer2[0]…
EncoderLayer4 (EncoderBlock)	[(None, 16, 16, 256), (None, 32, 32, 256)]	885,760	EncoderLayer3[0]…
EncodingSpace (EncoderBlock)	(None, 16, 16, 512)	3,540,992	EncoderLayer4[0]…
DecoderLayer1 (DecoderBlock)	(None, 32, 32, 256)	2,953,984	EncodingSpace[0]… EncoderLayer4[0]…
DecoderLayer2 (DecoderBlock)	(None, 64, 64, 128)	739,712	DecoderLayer1[0]… EncoderLayer3[0]…
DecoderLayer3 (DecoderBlock)	(None, 128, 128, 64)	185,536	DecoderLayer2[0]… EncoderLayer2[0]…
DecoderLayer4 (DecoderBlock)	(None, 256, 256, 32)	46,688	DecoderLayer3[0]… EncoderLayer1[0]…
OutputMap (Conv2D)	(None, 256, 256, 1)	33	DecoderLayer4[0]…

Total params: 8,640,109 (32.96 MB)
Trainable params: 8,635,303 (32.94 MB)
Non-trainable params: 4,806 (18.77 KB)

Figure 4: Model summary of the U-Net architecture used in the road-segmentation pipeline, showing the main encoder–decoder blocks, output tensor shapes, and parameter counts.

The training progress from the final epochs is shown below. The log indicates that the network was trained for 25 epochs and reached stable performance with consistent training accuracy and improving overlap-based segmentation metrics. The present training run was limited to 25 epochs because dedicated GPU access is currently unavailable. The experiments are being executed on Kaggle and Google Colab, both of which provide only time-limited and resource-limited GPU sessions. As a result, long uninterrupted training schedules are difficult to maintain, so the current results are reported for a 25-epoch run under these practical compute constraints.

Epoch	Accuracy	Dice	IoU	Val Dice	Val IoU
21	0.9754	0.3690	0.2827	0.3237	0.2402
22	0.9762	0.3678	0.2818	0.3124	0.2314
23	0.9766	0.3692	0.2840	0.3297	0.2449
24	0.9767	0.3817	0.2956	0.3110	0.2290
25	0.9777	0.3944	0.3084	0.3361	0.2503

Table 2: Training summary for the final five epochs of the road-segmentation experiment.And it's corresponding Epoch, Accuracy, Dice Coefficient, Intersection over Union and Dice vs. IoU

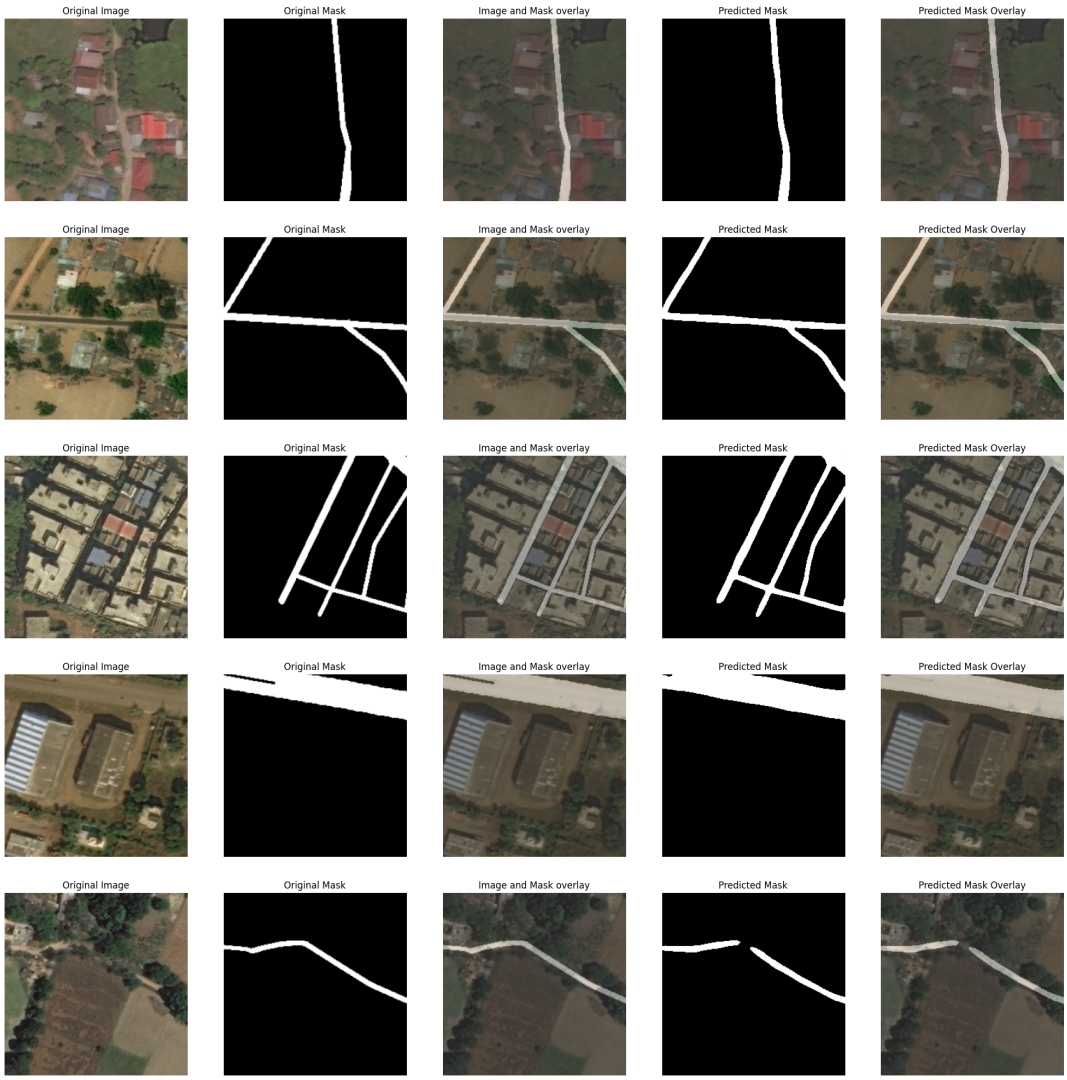


Figure 5: Qualitative road-segmentation results for multiple samples. Each panel compares the input satellite image, ground-truth mask, overlay visualization, predicted mask, and predicted mask overlay.

6 Future Work

The future direction of this project is centered on the final objective of generating road masks from satellite imagery using SAM. After developing a practical foundation through the U-Net-based baseline, the next stage of the work will focus on adapting SAM more directly to the road-extraction task and evaluating how effectively it can replace or improve upon the baseline segmentation approach.

A major expected benefit of this transition is that SAM may provide stronger generalization and reduced dependence on task-specific model design, while still preserving the fine and elongated

structure of roads. Building on the understanding gained from the U-Net stage, the SAM-based extension will aim to produce cleaner road masks, reduce fragmentation, and improve robustness in visually complex scenes containing shadows, buildings, vegetation, and low-contrast road segments.

In the longer term, these improved masks are expected to support more reliable intersection detection, connectivity estimation, and graph construction. In this way, the future work does not simply extend the current baseline; rather, it moves the project toward its actual end goal: a SAM-based road-segmentation pipeline grounded in the practical knowledge gained from earlier deep-learning and image-processing experiments.

References

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